

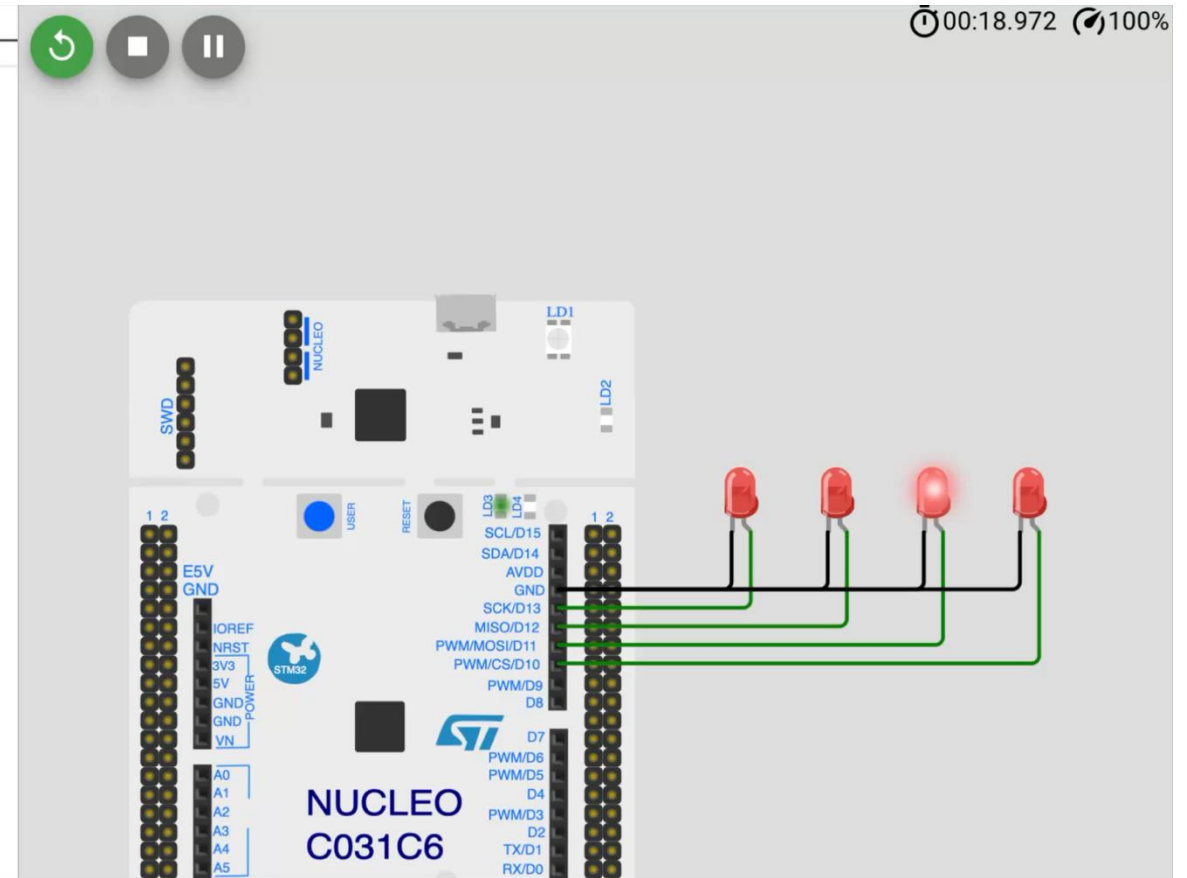
Digital sampling, ADC and DAC

Yuning You

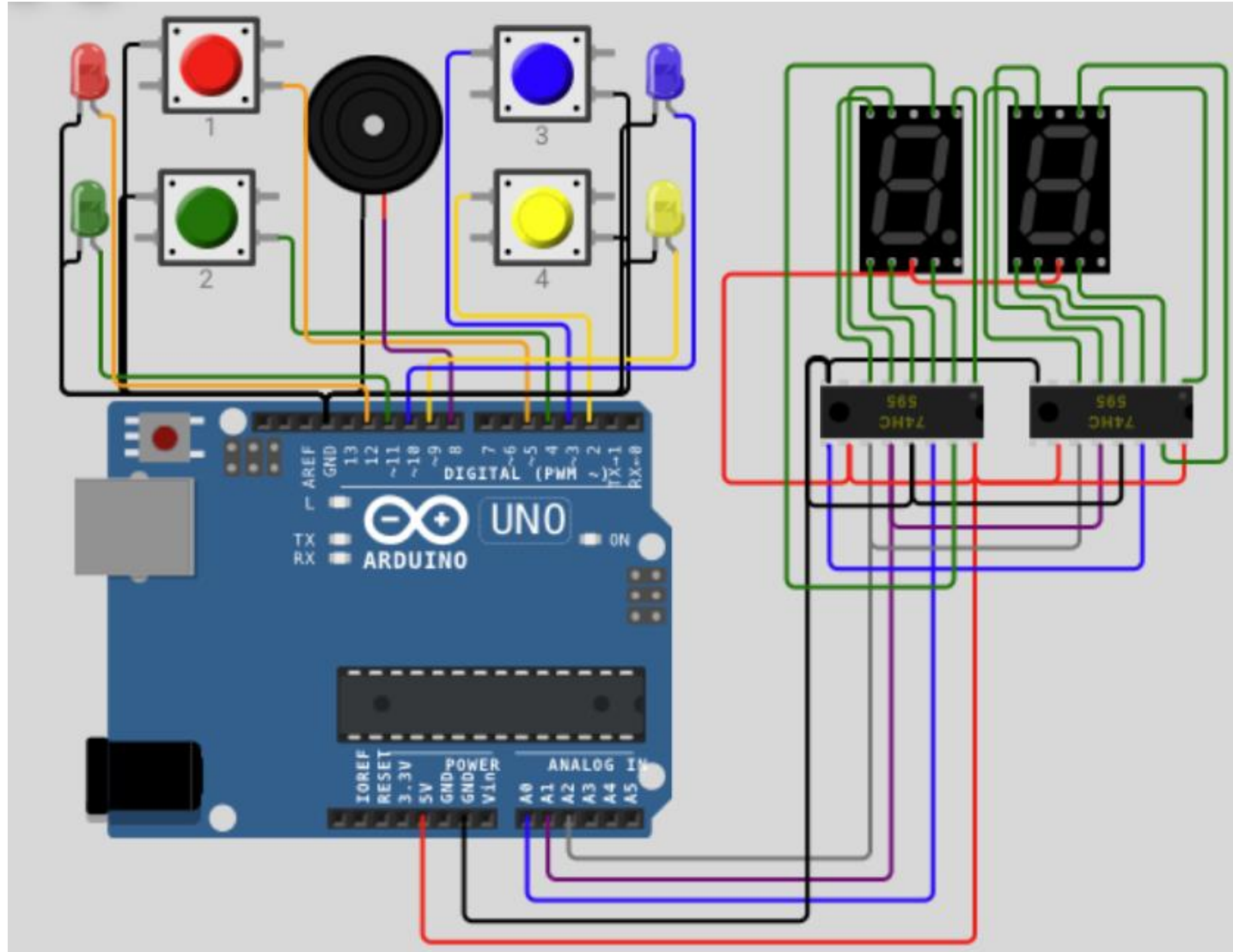
School of Science and Engineering
The Chinese University of Hong Kong, Shenzhen

Recap: Running LEDs

```
1 // 4-bit binary counter using LEDs (each LED toggles at different rates)
2
3 int ledPin[] = {13, 12, 11, 10};           // LED pins: MSB → LSB
4 int ledDelay[] = {8000, 4000, 2000, 1000}; // toggle interval (ms)
5 unsigned long lastChangeTime[4];          // store last toggle time of each LED
6 int ledState[4] = {LOW, LOW, LOW, LOW};   // store current state (LOW/HIGH)
7 unsigned long currTime;                   // current time variable
8
9 void setup() {
10     currTime = 0;                          // initialize timer
11     for (int i = 0; i < 4; i++) {
12         pinMode(ledPin[i], OUTPUT);        // set pin as output
13         digitalWrite(ledPin[i], ledState[i]); // turn LED off initially
14         lastChangeTime[i] = millis();      // record initial time
15     }
16 }
17
18 void loop() {
19     currTime = millis();                   // get current time
20
21     for (int i = 0; i < 4; i++) {
22         // check all LEDs
23         if (currTime - lastChangeTime[i] >= ledDelay[i]) {
24             // time to toggle the LED
25             ledState[i] = !ledState[i];    // flip LED state
26             digitalWrite(ledPin[i], ledState[i]); // apply new state
27             lastChangeTime[i] = currTime;  // reset timer for that LED
28         }
29     }
30 }
```



Recap: Simon game



ADC/DAC Real-World Applications

Audio Systems

Recording and playback of music, voice, and sound effects. Requires sampling rates of **44.1 kHz** or higher to capture the full audible spectrum.

FREQ: 20Hz-20kHz

Sensor Networks

Precise data acquisition for environmental monitoring. **Temperature, pressure, and humidity** sensors rely on accurate ADCs to digitize physical phenomena.

PRECISION: HIGH

Control Systems

Critical feedback loops in **robotics and automation**. ADCs measure state, and DACs drive actuators to maintain system stability and performance.

LATENCY: LOW

The Signal Chain

01. Input Path

Physical Phenomenon → Transducer → ADC →
Microcontroller → Digital Processing

02. Output Path

Microcontroller → DAC → Amplification →
Actuator → Physical Effect

03. Feedback Loops

Enables closed-loop control systems with real-time responsiveness by connecting Output back to Input.

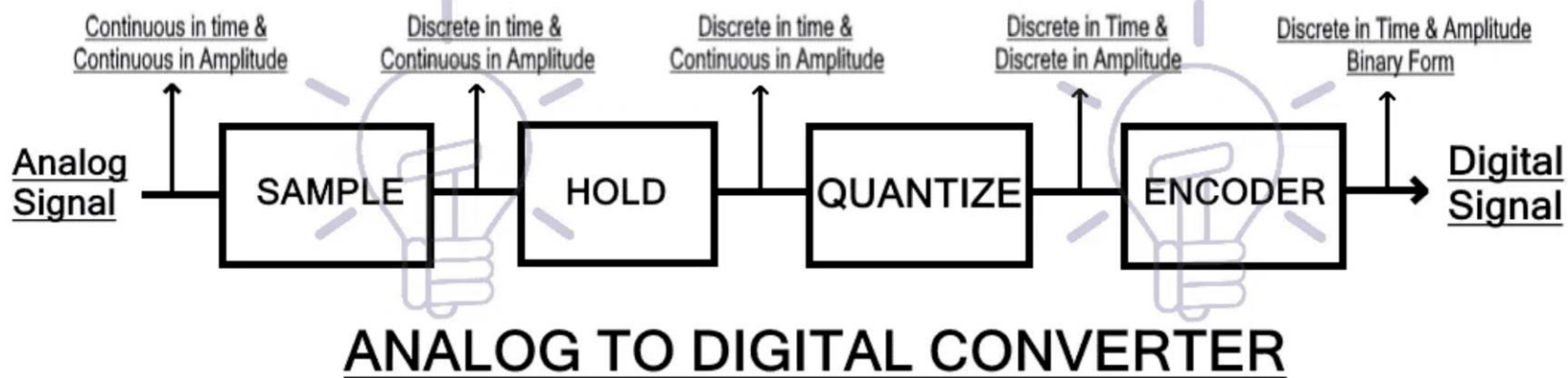


Figure: Analog to Digital Conversion Process

Essential Terminology

Measurand

The real-world physical signal or quantity of interest that is to be measured.

Examples: Sound pressure, temperature, force, acceleration, light intensity.

Transducer

A device that converts the physical measurand into an electrical signal (voltage or current) suitable for processing.

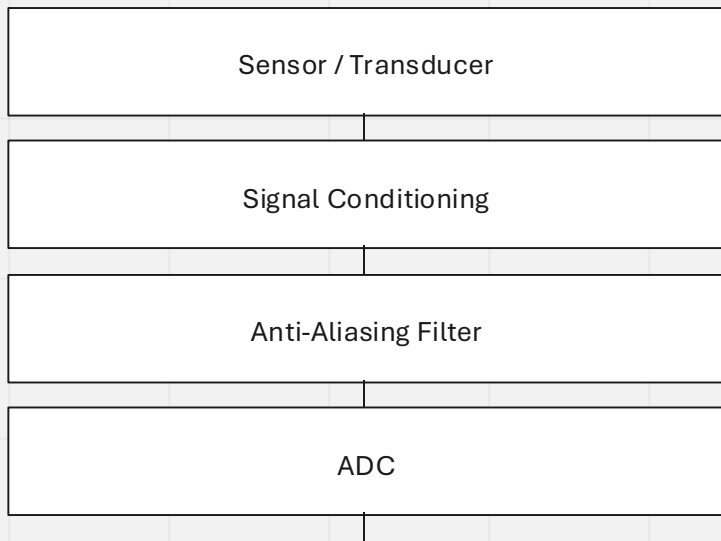
Examples: Microphone, thermistor, strain gauge, photodiode.

Sampling

The process of periodically measuring a continuous analog signal at fixed time intervals to create a discrete sequence of digital values.

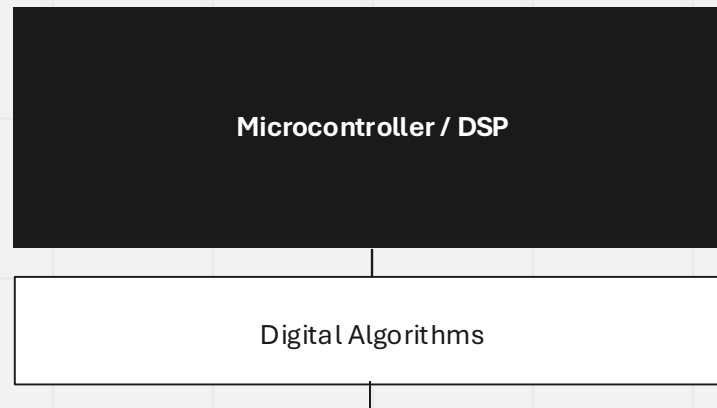
System Architecture

Input Subsystem



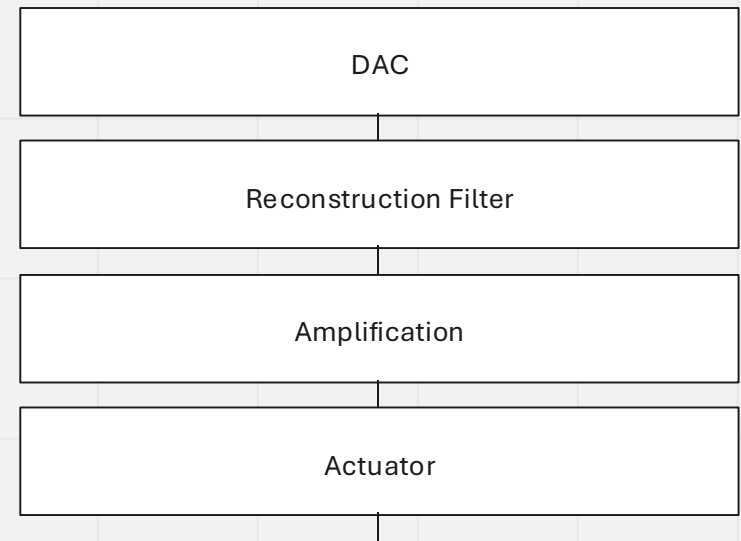
Captures physical phenomena and converts them into digital data. Critical for signal fidelity and noise rejection.

Processing



Executes control logic, filtering, and analysis. Must meet real-time constraints and computational efficiency requirements.

Output Subsystem



Converts digital commands back to analog signals to drive physical actuators and close the control loop.

Four Fundamental Constraints

Amplitude Resolution

Constraint 01

The smallest change in input signal that can be distinguished. It is fundamentally limited by the ADC's bit depth and system noise floor.

LIMIT: BIT DEPTH & NOISE

Amplitude Range

Constraint 02

The absolute minimum and maximum input values that can be measured. Signals outside this window are clipped or distorted.

LIMIT: SENSOR SPECS

Time Constraints

Constraints 03 & 04

Time Quantization: The sampling period ($1/f_s$).

Time Interval: The total duration of data collection.

LIMIT: SAMPLING RATE

What is Digital Sampling?

Continuous to Discrete

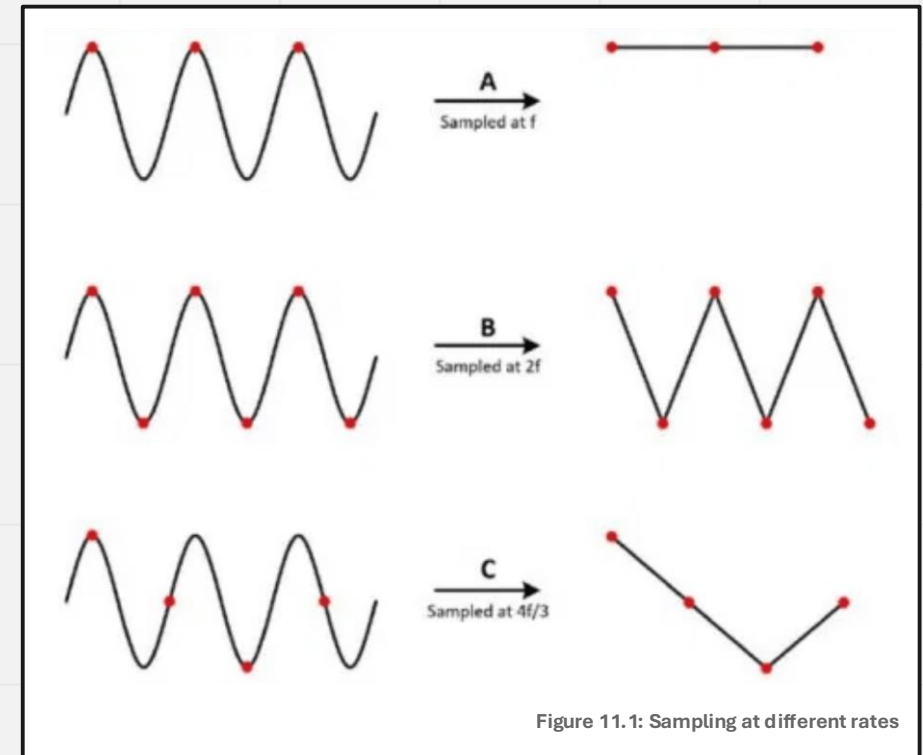
Digital sampling converts a continuous analog signal into a sequence of discrete digital values measured at fixed time intervals.

Sampling Rate (f_s)

The frequency at which we measure the signal, typically expressed in samples per second (Hz). It determines the temporal resolution.

Fundamental Operation

Sampling is the core process that enables digital signal processing, allowing analog reality to be stored and analyzed by computers.



Sampling Rate & Frequency

f_s

Sampling Rate

The number of samples taken per second, measured in Hertz (Hz). It defines the temporal resolution of the digital signal.

Example

44.1 kHz = 44,100 samples/sec

$1/f_s$

Time Quantization

The time interval between consecutive samples. A higher rate results in a smaller interval, capturing faster signal changes.

Example

$1 / 44,100 \approx 22.7$ microseconds

Scale

System Balance

The rate must be chosen to capture all relevant information while balancing memory usage and computational load.

Trade-off

Higher Rate = More Detail + More Data

The Nyquist Theorem

Sampling Condition

$$f_s \geq 2 \times f_{\max}$$

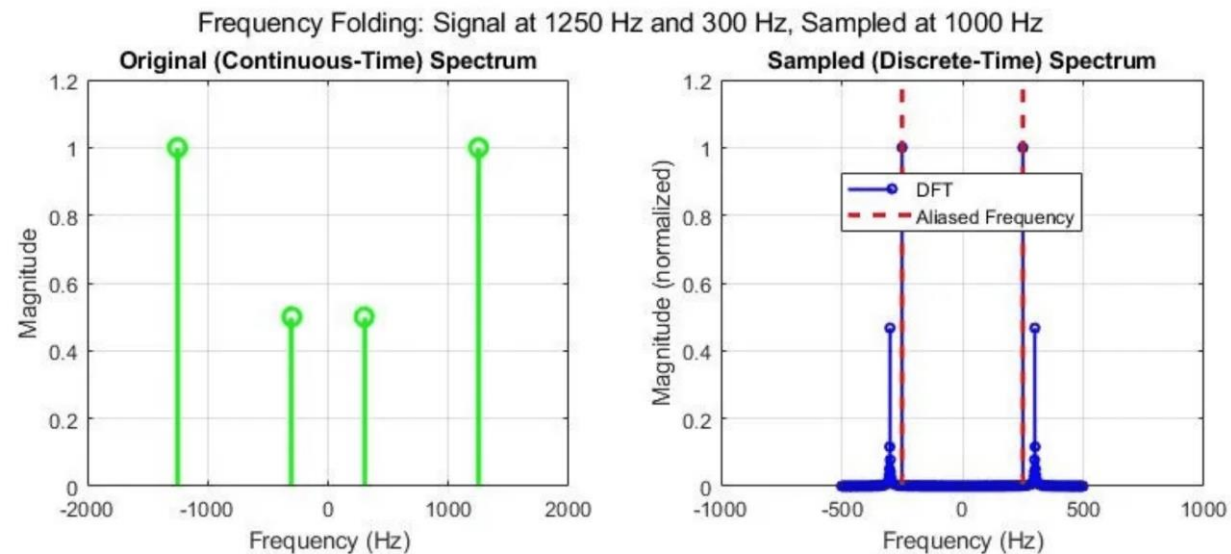
Perfect Reconstruction

A continuous signal can be perfectly reconstructed from samples if the sampling rate is at least twice the maximum signal frequency.

Mathematical Foundation

This theorem serves as the absolute mathematical foundation for all digital signal processing.

Consequence of Violation



The Nyquist Frequency

$$f_s/2$$

The Theoretical Limit

Definition

Also known as the folding frequency, it is exactly half the sampling rate. It represents the highest frequency component that can be accurately represented in the sampled data.

The Danger Zone

Any signal frequency above this limit will not be lost, but will instead "fold over" and appear as a lower frequency alias, corrupting the data.

Design Implication

This physical limit mandates the use of analog low-pass filters (Anti-Aliasing Filters) to strictly remove all content above $f_s/2$ before digitization.

Aliasing Artifacts

Critical Signal Error

Frequency Impersonation

When a signal exceeds the Nyquist limit ($f_s/2$), it "folds over" and appears as a lower frequency alias in the digital domain.

Irreversible Damage

Once aliasing occurs, the original signal is lost forever. No amount of digital post-processing can distinguish the alias from a real signal.

System Failure

Manifests as metallic distortion in audio, Moiré patterns in imaging, or false readings in control systems, potentially causing system instability.

ERROR

Real-World Aliasing

Temporal Domain

Wagon Wheel Effect

A classic stroboscopic effect where spoked wheels appear to rotate backwards or stand still because the camera's frame rate is too slow to capture the motion accurately.

Context: Video & Cinema

Spatial Domain

Moiré Patterns

Strange wavy interference patterns that appear when a fine grid (like a striped shirt) is photographed by a digital sensor with a fixed pixel grid.

Context: Photography & Print

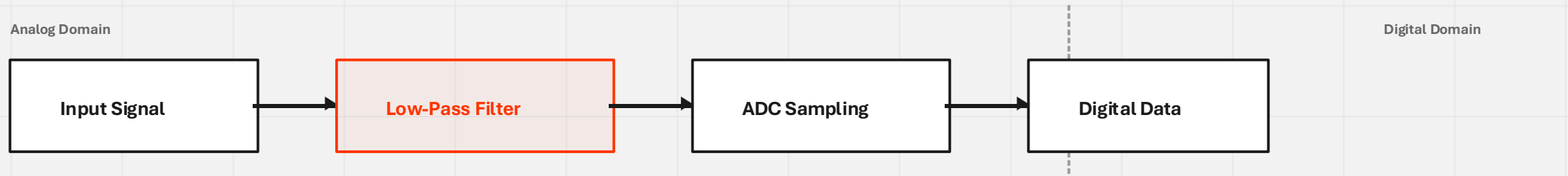
Spectral Domain

Audio Artifacts

High-frequency sounds folding back into the audible range as dissonant, metallic noises. This destroys signal fidelity and cannot be filtered out post-sampling.

Context: Digital Audio

Anti-Aliasing Filters



Irreversible Error

Aliasing cannot be fixed digitally. Once high frequencies fold into the baseband, they are indistinguishable from real signals. **Prevention is mandatory.**

Analog Solution

An analog Low-Pass Filter (LPF) must be placed before the ADC input to attenuate frequencies above the Nyquist limit.

Cutoff Frequency

The filter's cutoff frequency (f_c) is set to satisfy the Nyquist condition:

$$f_c \leq f_s / 2$$

Oversampling Benefits

Definition Methodology	Sampling at a frequency significantly higher than the Nyquist rate. $f_s \gg 2 \times f_{max}$
SNR Gain Signal Quality	Spreads quantization noise over a wider bandwidth, reducing the noise floor in the signal band. +3dB SNR per Octave of Oversampling
Filtering Hardware Design	Increases the gap between the signal and its aliases, allowing for the use of simpler, lower-order analog anti-aliasing filters. Relaxed Analog Complexity

Practical Constraints

01

System Noise

Real-world signals are never pure. Thermal noise, electromagnetic interference (EMI), and quantization noise all reduce the Effective Number of Bits (ENOB).

Impact

Limits usable resolution regardless of ADC bit depth.

02

Clock Jitter

Variations in the timing of the sampling clock cause errors in the sampled voltage, particularly for high-frequency input signals.

Impact

Reduces Signal-to-Noise Ratio (SNR) at high frequencies.

03

Resource Budget

High-speed, high-resolution sampling requires significant power, expensive components, and heavy computational load for processing.

Impact

Forces trade-offs between performance and cost/power.

Key Principles Summary

01

The Golden Rule

The Nyquist Theorem ($f_s \geq 2f_{\max}$) is the absolute mathematical condition for perfect signal reconstruction. Violating it destroys data integrity.

02

The Gatekeeper

Analog anti-aliasing filters are mandatory. They must strictly attenuate all frequencies above the Nyquist limit before the signal reaches the ADC.

03

The Trade-off

Practical design involves balancing sampling rate, filter complexity, and noise performance. Oversampling is a powerful technique to relax analog requirements.

Analog-to-Digital Converter

A
D
C

The Digital Gateway

The Bridge

The ADC serves as the critical interface that connects the continuous, infinite-resolution analog world to the discrete, binary digital domain.

The Translator

It translates physical measurements (voltage, current) into binary codes that microcontrollers and DSPs can process, store, and analyze.

System Fidelity

The accuracy, speed, and resolution of the ADC fundamentally define the upper performance limit of the entire digital signal processing system.

ADC Specifications I

n

Resolution

The number of bits used to represent the analog signal. It defines the granularity of the digital output code.

Typical Values

8-bit, 10-bit, 12-bit, 16-bit

2n

Precision

The total number of discrete levels or steps that the ADC can distinguish across its full range.

Example (10-bit)

$$2^{10} = 1024 \text{ levels}$$

V_{in}

Range

The span of input voltages the ADC can convert, typically defined by reference voltages ($V_{\text{ref+}}$ and $V_{\text{ref-}}$).

Constraint

$$0V \leq V_{\text{in}} \leq V_{\text{ref}}$$

Resolution & Accuracy

Voltage Resolution (LSB)

$$V_{\text{LSB}} = V_{\text{range}} / 2^n$$

Voltage Resolution

The smallest change in analog voltage that results in a change in the digital output code. It represents the "step size" of the staircase approximation.

Accuracy Metric

Accuracy is distinct from resolution. While resolution defines precision (number of bits), **accuracy** defines how close the digital output is to the true analog value, accounting for all error sources.

Practical Resolution

Bit Depth	Levels (2 ⁿ)	Step Size (5V Ref)	Typical Application
8-bit	256	19.53 mV	Basic microcontrollers, simple joysticks, battery monitoring.
12-bit	4,096	1.22 mV	Industrial sensors, standard data acquisition, motor control.
16-bit	65,536	0.076 mV	High-fidelity audio, scientific instrumentation, medical devices.

** LSB values calculated assuming a 0-5V input range.*

Quantization Error

± 0.5
LSB

Maximum Theoretical Error

The Rounding Effect

Since an ADC has a finite number of steps, it must round any continuous input voltage to the nearest available digital code.

The Error Limit

Ideally, the difference between the actual input voltage and the digitized value will never exceed half the width of one step (Least Significant Bit).

Inherent Nature

This error is not a malfunction but a fundamental consequence of digitization. It cannot be eliminated, only reduced by increasing resolution (n).

Quantization Noise & SNR

Error as Noise

Quantization error is often modeled as additive white noise. It is assumed to be uncorrelated with the signal and uniformly distributed between -0.5 LSB and +0.5 LSB.

The Limit

This "noise floor" is intrinsic to the digital conversion process. No matter how clean the analog signal is, the digital representation will always have this uncertainty.

Theoretical SNR (dB)

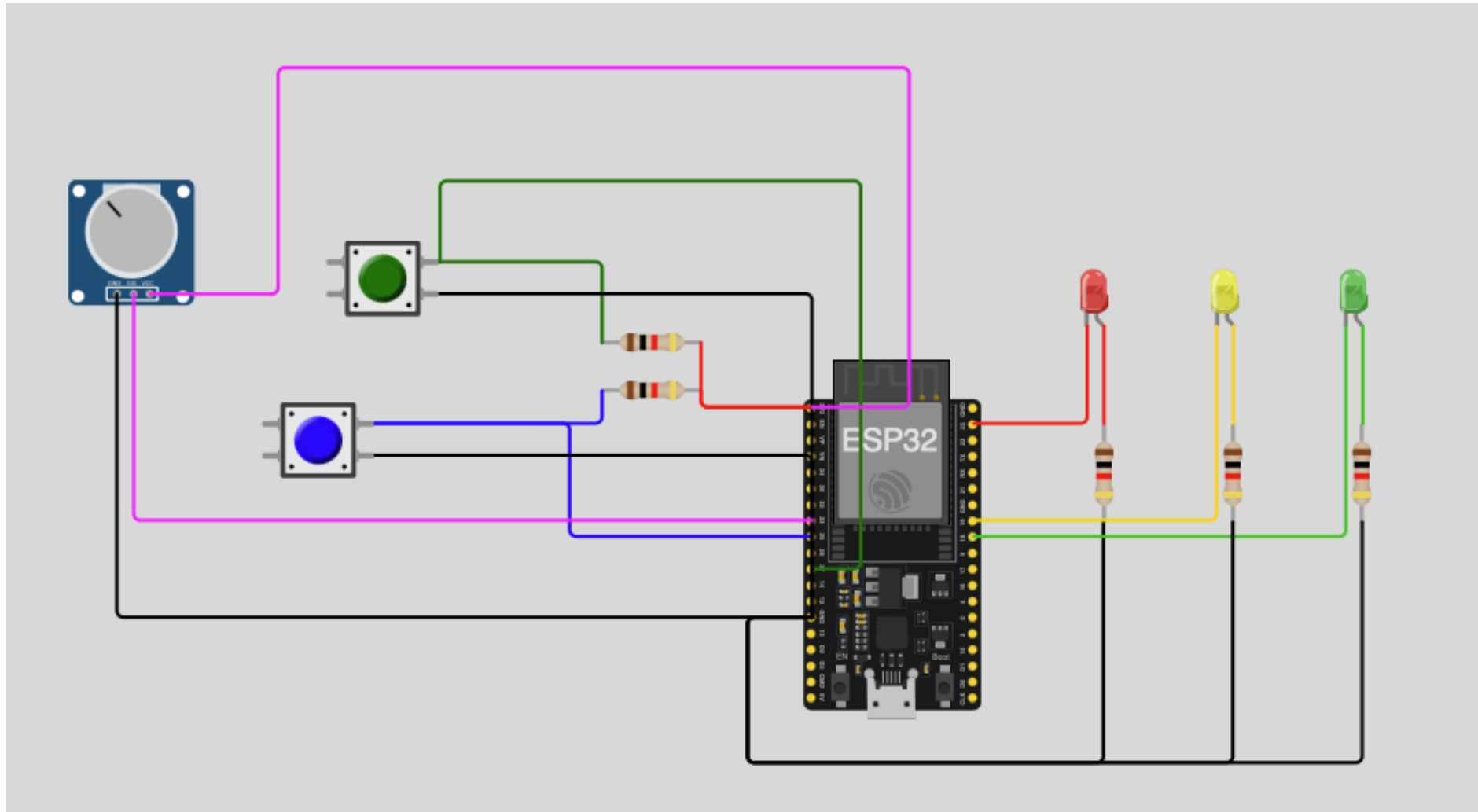
$$\text{SNR} = 6.02n + 1.76$$

6dB

The Golden Rule:

For every additional bit of resolution (n), the Signal-to-Noise Ratio improves by approximately **6 dB**.

Virtual lab: ADC



Virtual lab: ADC

- Explain the logic of the simulation program

Virtual lab: ADC

- Modify the logic:
 - If $1000 < \text{ADC value} < 2000$: Light yellow LED
 - If $2000 < \text{ADC value} < 3500$: Light red LED

ADC Conversion Process

01

Sample & Hold

The input switch closes briefly to charge a capacitor, "freezing" the rapidly changing analog voltage at a specific instant so it can be measured.

→ 02

Quantization

The held voltage is compared against reference levels. It is rounded to the nearest discrete value available in the ADC's resolution grid.

→ 03

Encoding

The quantized level is converted into a binary number (e.g., 10110100) and output to the digital bus for processing by the microcontroller.

Major ADC Architectures

SAR

Medium Speed / High Res

Successive Approximation Register: The standard for general-purpose microcontrollers. Balances resolution (8-18 bits) with sample rates up to 5 MSPS.

Flash

Ultra High Speed

Uses a massive array of parallel comparators for instant conversion. Extremely fast (>1 GSPS) but limited to low resolution (6-8 bits) due to complexity.

Sigma-Delta

High Precision

Σ - Δ : Uses oversampling and noise shaping to achieve very high resolution (16-24+ bits) at the cost of lower speed and higher latency.

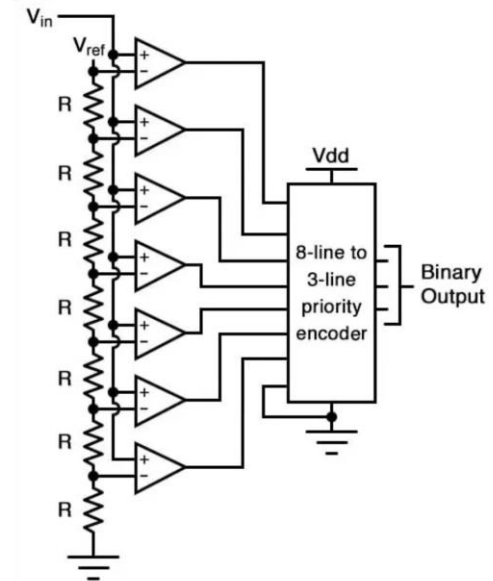


Figure : Flash ADC Parallel Architecture

Specialized Architectures

High Accuracy

Dual-Slope

An integrating architecture that charges and discharges a capacitor. It is extremely slow but offers exceptional noise immunity by averaging out high-frequency noise.

Primary Use

Digital Multimeters (DMMs)

Simplicity

Ramp

Uses a counter, DAC, and comparator to ramp up voltage until it matches the input. Simple and low cost, but conversion time varies with input signal magnitude.

Primary Use

Low-cost, low-speed sensing

Throughput

Pipeline

Divides the conversion into multiple stages, processing bits in parallel. Offers high resolution at very high speeds, but introduces data latency (pipeline delay).

Primary Use

Video, Ultrasound, Comms

SAR ADC Architecture

Binary Search Mechanism

Successive Approximation Register (SAR) ADCs work like a balance scale, testing each bit from MSB to LSB to converge on the input voltage value.

Conversion Efficiency

It requires exactly **n clock cycles** to convert an n-bit value, making the conversion time deterministic and efficient.

Industry Standard

Due to its excellent balance of speed, resolution, and power consumption, SAR is the dominant architecture in modern microcontrollers.

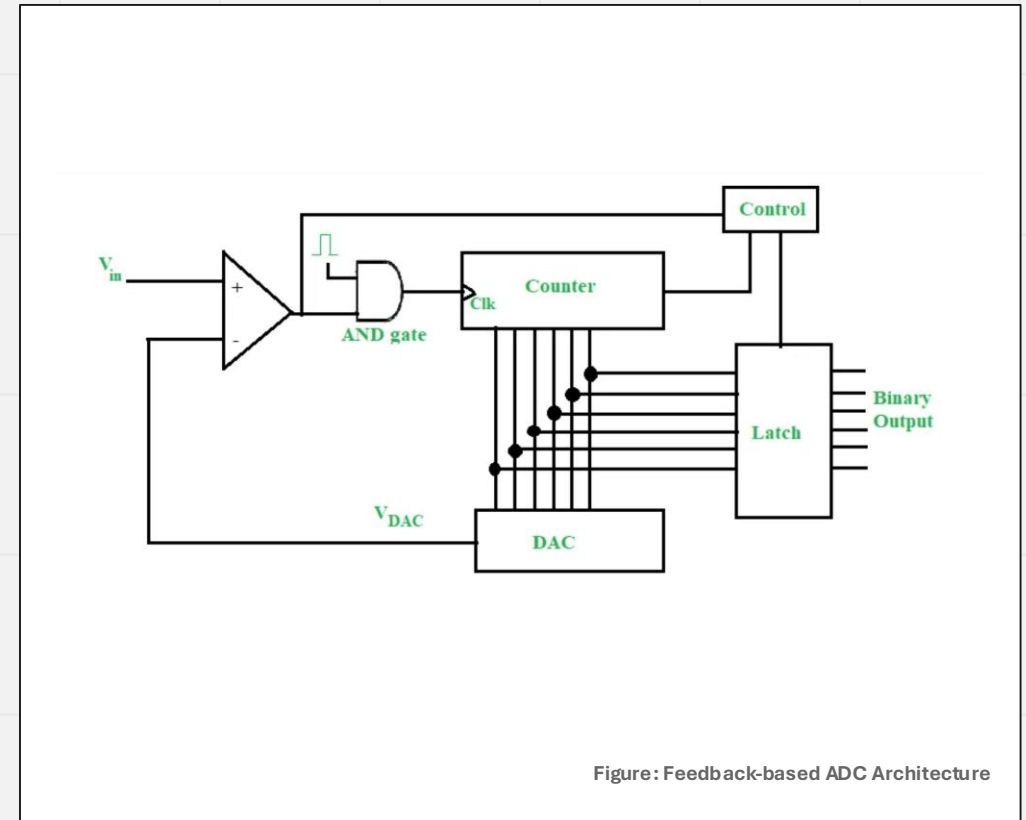


Figure: Feedback-based ADC Architecture

ADC Timing Specs

Total Cycle Time

Acquisition

Conversion

Acquisition Time

The time required for the internal Sample & Hold capacitor to charge to the input voltage level. Depends heavily on source impedance.

Conversion Time

The time taken by the ADC circuitry to process the held voltage and generate the digital code. This is fixed by the ADC architecture.

Total Time

The sum of acquisition and conversion times. This defines the maximum possible sampling rate of the system.

$$\text{Max } f_s = 1 / (t_{\text{acq}} + t_{\text{conv}})$$

Source Impedance Impact

R_{in}

Voltage Divider Effect

High source impedance creates a voltage divider with the ADC's internal input resistance. This causes the voltage at the ADC pin to be lower than the actual sensor voltage, introducing a gain error.

RC

Acquisition Lag

The source resistance (R) and the ADC's sampling capacitor (C) form an RC circuit. A high R increases the time constant ($\tau = R \times C$), requiring a longer acquisition time to fully charge the capacitor.

OP

The Buffer Solution

To fix both issues, an Operational Amplifier (Op-Amp) buffer is placed between the sensor and the ADC. It provides high input impedance for the sensor and low output impedance to drive the ADC quickly.

ADC Nonlinearity

An ideal ADC has a perfectly linear transfer function with uniform step widths. Real-world components introduce deviations that distort the digital output.

DNL

Differential Nonlinearity

The difference between the actual step width and the ideal 1 LSB width. It measures the "jitter" in step sizes.

Critical Failure: If $DNL < -1 \text{ LSB}$, the ADC becomes non-monotonic and "missing codes" occur (some values are never output).

INL

Integral Nonlinearity

The maximum deviation of the actual transfer function from a straight line (drawn through endpoints or best fit).

Critical Failure: Represents the cumulative error across the entire range, manifesting as "curvature" or global distortion.

Real-World Applications

Thermal

Temperature Sensing

Industrial process control and environmental monitoring rely on ADCs to digitize small voltage changes from thermal sensors. Precision is key here.

Typical Sensors

Thermocouples, RTDs, Thermistors

Kinetic

Pressure & Force

From automotive manifold pressure (MAP) to digital scales, physical force is converted to voltage via strain gauges and then digitized for the ECU or display.

Typical Sensors

Piezoelectric, Strain Gauges, Load Cells

Optical

Light Intensity

Cameras, ambient light sensors, and optical communications convert photon energy into current, which is transimpedance-amplified and then sampled.

Typical Sensors

Photodiodes, Phototransistors, CCDs

Design Selection

The Fundamental Trade-off

Speed vs. Resolution

You cannot maximize both simultaneously without exponential cost. **High Speed** (Flash) sacrifices bits. **High Resolution** (Sigma-Delta) sacrifices bandwidth. SAR offers the middle ground for most control systems.

System Constraint

Power Budget

Power consumption scales linearly with sampling rate and exponentially with resolution. For battery-powered IoT devices, selecting the **minimum viable specification** is critical to extend operational life.

Environmental Factor

Noise Immunity

In electrically noisy industrial environments, **differential inputs** are mandatory to reject common-mode noise. Single-ended inputs should only be used in clean, short-trace PCB environments.

Digital-to-Analog Converter

D
A
C

The Analog Output

The Inverse Process

A Digital-to-Analog Converter (DAC) performs the exact inverse operation of an ADC, transforming binary digital codes back into continuous analog voltages or currents.

Signal Reconstruction

It reconstructs the original signal by holding discrete voltage levels for specific time intervals, approximating the smooth analog waveform from a series of steps.

Physical Interface

DACs are the essential link that allows digital systems to interact with the real world, driving speakers, motors, displays, and communication transmitters.

DAC Specifications I

n

Resolution

The number of bits in the digital input code. It determines how finely the output range can be divided.

Common Values

8-bit, 12-bit, 16-bit, 24-bit

2n

Precision

The total number of unique analog voltage levels the DAC can generate. Higher precision means smoother output curves.

Example (12-bit)

$$2^{12} = 4096 \text{ levels}$$

V_{out}

Range

The span of possible output voltages, typically from 0V to a reference voltage (V_{ref}).

Constraint

$$0V \leq V_{\text{out}} < V_{\text{ref}}$$

DAC Specifications II

Settling Time

The time required for the output voltage to settle within a specified error band (e.g., ± 0.5 LSB) after a code change. This is the primary factor limiting the DAC's maximum update rate.

Critical For

High-speed waveform generation

Glitch Energy

Transient voltage spikes that occur during major carry transitions (e.g., 0111 to 1000) due to internal switch timing mismatches. It is measured as the area under the voltage-time curve.

Unit

nV-s (Nanovolt-seconds)

Output Impedance

The internal resistance seen looking into the DAC output. High output impedance limits the ability to drive loads directly, often necessitating an external Op-Amp buffer.

Requirement

Buffer required for resistive loads

Practical DAC Resolution

Bit Depth	Levels (2 ⁿ)	Step Size (3.3V Ref)	Typical Application
8-bit	256	12.89 mV	LED brightness control, simple bias voltage generation.
12-bit	4,096	0.80 mV	Industrial control loops, motor speed reference, function generators.
16-bit	65,536	0.05 mV	High-fidelity audio playback, precision medical instrumentation.

** LSB values calculated assuming a 0-3.3V output range.*

Binary-Weighted DAC

Scaling Principle

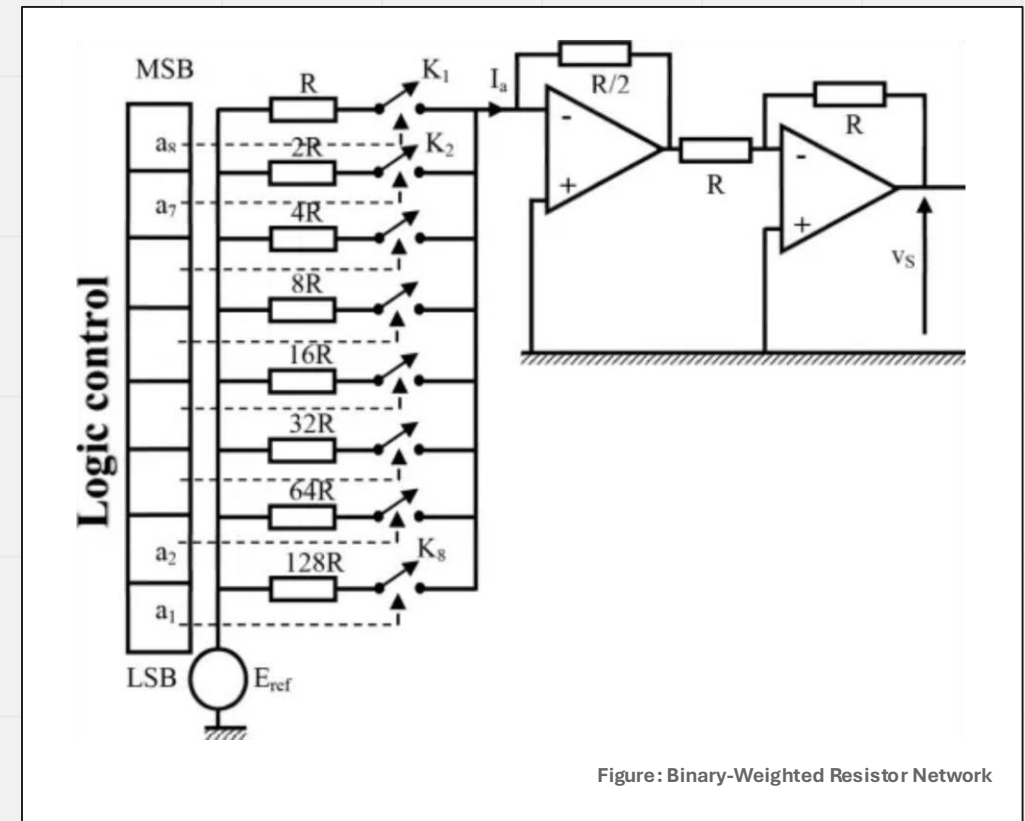
This architecture uses a bank of resistors in parallel, where each resistor's value is inversely proportional to the binary weight of its bit (R , $2R$, $4R$, $8R$...).

Summing Operation

An Operational Amplifier in an inverting summing configuration adds the currents from the active bits to produce an output voltage proportional to the binary code.

The Fatal Flaw

It requires precise resistor ratios that grow exponentially. For a 12-bit DAC, the MSB resistor must be **4096 times smaller** than the LSB resistor, which is impossible to manufacture accurately on a chip.



2-Bit Calculation

Reference Voltage

5.0 V

Input Code (Binary)

10

MSB = 1, LSB = 0

Formula

$$V_{\text{out}} = V_{\text{ref}} \times (b_1 2^{-1} + b_0 2^{-2})$$

Substitute

$$V_{\text{out}} = 5V \times (1 \times 0.5 + 0 \times 0.25)$$

Simplify

$$V_{\text{out}} = 5V \times (0.5)$$

Output Voltage

2.5 V

R-2R Ladder DAC

Precision Matching

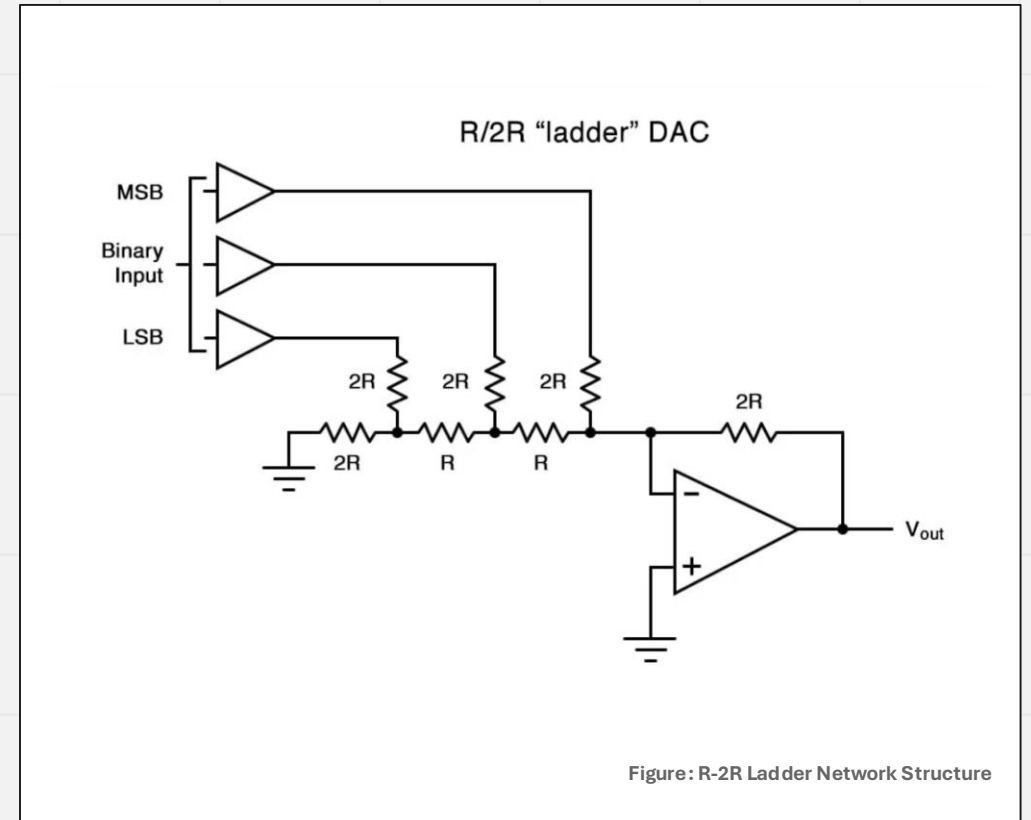
Unlike binary-weighted DACs, this architecture uses only two resistor values (**R and 2R**). This makes it much easier to manufacture precise, matched components on a silicon die.

Constant Impedance

The output impedance remains constant (equal to R) regardless of the digital input code. This simplifies the design of the output buffer amplifier.

Scalability

Increasing resolution is simple: just add more "rungs" to the ladder. It avoids the massive resistor ratios required by binary-weighted designs.



The R-2R Ladder

Component Simplicity

Unlike binary-weighted DACs that require a vast range of resistor values, the R-2R ladder uses only two distinct values: R and 2R. This makes precision matching on integrated circuits feasible and cost-effective.

Advantage: Manufacturability

2

Current Division

The ladder structure naturally divides the current by exactly two at each node, regardless of the number of bits. This creates the required binary weighting ($1/2$, $1/4$, $1/8$...) purely through topology.

Advantage: Intrinsic Accuracy

÷

Modular Scalability

The structure is repetitive. To increase resolution from 8-bit to 12-bit, you simply add four more identical R-2R sections to the ladder without changing the existing component values.

Advantage: Extensibility

∞

DAC Architectures

Resistor Network

Uses voltage division across a string of resistors or an R-2R ladder. It is simple, fast, and monotonic, but component matching limits resolution to about 12-14 bits.

Best For

General Purpose, Video

Switched Capacitor

Uses charge redistribution on capacitor arrays instead of resistors. Capacitors are easier to match in CMOS processes, making this ideal for integrated circuits.

Best For

Integrated Microcontrollers

Sigma Delta

Uses 1-bit DAC with massive oversampling and noise shaping. It trades speed for exceptional linearity and resolution (up to 24+ bits).

Best For

High-Fidelity Audio

Output Architecture

Voltage Output

These DACs include an internal output amplifier. They directly produce a voltage (e.g., 0-5V) and are easy to interface with.

- + Simple implementation
- Slower settling time
- Limited drive capability

Current Output

These DACs output a current proportional to the code. They require an external Op-Amp (I-to-V converter) to generate a voltage.

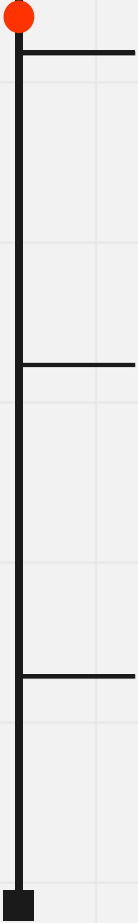
- + Extremely high speed
- + Wide bandwidth
- Requires external components

LOAD

The Buffering Rule

DAC outputs typically have high impedance or limited current. To drive low-impedance loads (like speakers or long cables) without voltage droop, a unity-gain Op-Amp buffer is mandatory.

Settling Time



01 Definition

The time elapsed from the application of a digital input code change until the output voltage enters and **remains within** a specified error band (typically ± 0.5 LSB) of the final value.

02 Composition

It consists of two phases: the **Slew Time** (limited by the maximum rate of voltage change) and the **Linear Settling Time** (where small-signal ringing and oscillations dampen out).

03 System Limit

Settling time sets the hard limit on the DAC's maximum update rate. If the DAC is updated faster than it can settle, the output waveform will be distorted and amplitude-attenuated.

DAC Glitches

The Cause: Timing Mismatch

Internal switches within the DAC do not open and close at the exact same instant. During a code change, some switches may turn on slightly before others turn off, creating a momentary invalid state.

Worst Case: Major Carry

011...1 → 100...0

The largest glitches occur at mid-scale transitions where every single bit flips. If the MSB turns on slower than the LSBs turn off, the output voltage can momentarily drop to near zero before recovering.

The Metric: Glitch Energy

nV-s

Because glitches are extremely fast transients, they are quantified by their "impulse area"—the product of the voltage spike amplitude and its duration (Nanovolt-seconds).

DAC Applications

Audio & Media

The most ubiquitous application. DACs convert stored digital audio files (MP3, FLAC) into analog electrical signals that drive headphones and speakers.

Examples

Smartphones, Sound Cards, Digital Synthesizers

Test & Measure

In laboratory equipment, high-speed DACs are used to generate precise waveforms (sine, square, arbitrary) for testing electronic circuits and systems.

Examples

Function Generators, Arbitrary Waveform Generators

Industrial Control

DACs act as the interface between digital controllers and the physical world, driving analog actuators that require variable voltage or current inputs.

Examples

Motor Speed Controllers, Valve Positioners, Laser Control

DAC Design Selection

Performance Trade-off

Speed vs. Resolution

Match the architecture to the signal bandwidth. Audio requires **high resolution** (16-24 bit) but moderate speed. Video and RF generation demand **high speed** (>100 MHz) but tolerate lower resolution (8-10 bit).

Control Requirement

Monotonicity

In closed-loop control systems, non-monotonic behavior (where output drops as input code increases) can cause **oscillation and instability**. Always verify the "Guaranteed Monotonic" spec for control applications.

Interface Constraint

Output Buffering

Most DACs cannot drive low-impedance loads directly. To drive cables, speakers, or actuators without voltage sag, a precision **Op-Amp buffer** is mandatory to isolate the DAC from the load.

System Complementarity

ADC

Analog → Digital

DAC

Digital → Analog

System Role

The Observer

DSP Core

System Role

The Actuator

Captures the physical world by discretizing time (sampling) and amplitude (quantization). Its primary battle is against noise and aliasing to preserve information fidelity.

Reconstructs the physical world by holding discrete levels to approximate a continuous signal. Its primary battle is against glitches and settling time to ensure smooth output.

Digital Audio Chain

01 Capture

Microphone

Transducer converts acoustic pressure waves into a weak analog voltage signal.

ADC

Samples the amplified signal at ≥ 44.1 kHz to capture the full audible spectrum (20Hz-20kHz).

02 Process

DSP / Storage

Digital Signal Processors perform mixing, EQ, and compression mathematically without adding analog noise.

Transmission

Data is stored (WAV, MP3) or transmitted (Bluetooth, Wi-Fi) with perfect fidelity.

03 Playback

DAC

Reconstructs the analog waveform. A reconstruction filter removes high-frequency quantization noise.

Speaker

Power amplifier drives the voice coil to produce sound waves.

Audio Specifications

Reconstruction Filter

The raw output of an audio DAC is a "staircase" wave. A steep analog low-pass filter is mandatory to smooth these steps and remove high-frequency imaging artifacts.

Target Cutoff

20 kHz

Dynamic Range

The ratio between the loudest undistorted signal and the noise floor. 16-bit CD audio provides 96 dB, while 24-bit studio audio offers a theoretical 144 dB.

Hi-Fi Standard

> 100 dB

Signal Purity

Total Harmonic Distortion + Noise (THD+N) measures the unwanted signals added by the converter. In high-fidelity audio, this must be vanishingly low to ensure transparency.

Target THD+N

< 0.005%

Sensor Acquisition



Transduction

The process begins with a transducer converting a physical parameter (temperature, pressure, light) into a raw electrical signal, often very weak (mV or μA).

Challenge

Low Amplitude, High Noise

Conditioning

The raw signal must be amplified to fill the ADC's dynamic range and filtered to remove noise and prevent aliasing. This is the most critical analog stage.

Requirement

Gain, Filtering, Isolation

Digitization

The conditioned signal is finally sampled by the ADC. The quality of the digital data is entirely dependent on the quality of the conditioning stage preceding it.

Goal

Max SNR, Min Distortion

Closed-Loop Control

Sense

01

The system measures the actual physical state (e.g., speed, temperature) using sensors. The ADC digitizes this feedback signal for the controller.

ADC



Think

02

The digital controller compares the measured value to the desired setpoint, calculates the error, and computes a correction using algorithms like PID.

CPU / DSP



Act

03

The calculated correction is converted back to an analog signal by the DAC to drive the actuator (motor, heater) and adjust the physical process.

DAC

Feedback Loop

Motor Control Loop

Step 01: Sense

Measure Speed

A tachometer generates a voltage proportional to the motor's actual speed. The ADC samples this voltage to provide digital feedback.

ADC + Sensor

Step 02: Process

Compute Error

The microcontroller compares the measured speed (from ADC) against the target setpoint. A PID algorithm calculates the necessary correction.

Microcontroller

Step 03: Act

Drive Motor

The DAC converts the digital correction value into an analog control voltage, which the power amplifier uses to adjust the motor's speed.

DAC + Actuator

← Closed Loop Feedback ←

Signal Conditioning



Amplification

Sensor outputs are often in the millivolt range. We must apply gain to map this small signal to the ADC's full-scale input range (e.g., 0 to 3.3V), maximizing the use of available **resolution bits**.



Filtering

An analog Low-Pass Filter (Anti-Aliasing Filter) is mandatory before the ADC input. It removes high-frequency noise and ensures no signal components exist above the **Nyquist frequency**.



Offset Adjustment

Many real-world signals are bipolar (swinging positive and negative), while modern microcontrollers use single-supply ADCs (0V to Vref). We add a DC bias voltage to **center the signal** within the valid range.

Noise Management

Quantization Noise

The inherent error caused by rounding continuous values to the nearest digital step. It appears as a fixed noise floor determined by bit depth.

Mitigation

Increase resolution (more bits) or use Oversampling techniques to spread noise energy across a wider bandwidth.

Thermal Noise

Random voltage fluctuations caused by thermal agitation of electrons in resistors and semiconductor components (Johnson-Nyquist noise).

Mitigation

Select low-noise, low-resistance components and limit the system bandwidth to the minimum required frequency.

EMI / Interference

External noise coupled from power lines, radio frequencies, or adjacent digital switching circuits.

Mitigation

Use proper shielding, differential signaling (balanced lines), and careful PCB layout with separate analog/digital grounds.

Design Trade-offs

Conflict 01

Resolution

Speed

Fundamental hardware limit. Increasing bit depth (precision) exponentially increases the time required for conversion. You cannot easily have both 24-bit precision and GHz sampling rates.

The Reality

Choose: High Fidelity (Audio) OR High Speed (RF/Video).

Conflict 02

Bandwidth

Data Load

To capture higher frequencies, you must sample faster (Nyquist). This linearly increases the data rate, demanding more storage, faster processors, and higher power consumption.

The Reality

Don't over-sample. Stick to Nyquist + margin.

Conflict 03

Performance

Cost

High-performance converters (low noise, high linearity, fast settling) require premium silicon processes and careful trimming, making them exponentially more expensive.

The Reality

Specify only the precision actually required by the physics.

Thank you!